

OMIP-119: A 36-Color Full-Spectrum Flow Cytometry Panel for Deep Immunophenotyping of Peripheral Blood and Ex Vivo Expanded Human T Cells

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Received: 5 August 2025 | **Revised:** 18 November 2025 | **Accepted:** 21 November 2025

ABSTRACT

The increasing uptake of adoptive CAR-T and TCR-T cell therapies into clinical practice has intensified the need for robust and detailed immunophenotyping of ex vivo expanded T cells. We have developed and optimized an extended 36-color full-spectrum flow cytometry panel suitable for in-depth immunophenotyping of both peripheral blood and ex vivo expanded human T cells. Our panel allows for analysis of CD4⁺ and CD8⁺ memory subpopulations (Tn/Tscm, Tcm, Tem, and Temra), Treg subsets (CD45RA⁺ and CD39⁺), and helper T cell subsets (Tfh, Th1, Th2, and Th17). Intracellular staining facilitates the evaluation of functional markers granzyme B (cytotoxicity) and Ki-67 (proliferation). Our panel allows detailed investigation of T cell activation using markers carrying a range of expression kinetics. Exhaustion and senescence features, known to correlate with resistance to immunotherapies, can also be thoroughly evaluated. TCR staining with pHLA-dextramer or anti-TCR antibody is also included to allow evaluation of TCR specificity and/or transduction efficiency. A PD-1 receptor occupancy component enables quantification of T cells bound to antibodies from immune checkpoint inhibitor (ICI) therapies. Overall, we have generated a fully optimized and fit-for-purpose extended T cell profiling panel with particular relevance to the field of immunotherapies including ICIs and adoptive T cell therapies.

1 | Background

Technological advances in the field of flow cytometry have pushed beyond conventional approaches limited by measuring only the peak emission wavelengths toward utilizing the entire fluorescent spectrum with full-spectrum flow cytometry. With an expanding selection of fluorophores covering previously underutilized parts of the spectrum, full-spectrum panels comprising as many as 50 colors have been developed on specialized cytometers [1]. Aside from extending the limits on the number of parameters, full-spectrum analyses can also deliver lower limits of detection and higher signal resolution [2].

Full-spectrum flow cytometry panels assembled for T-cell immunophenotyping have primarily focused on characterizing the steady-state phenotype of T cells derived from peripheral blood

[3, 4]. However, the increasing uptake of adoptive T cell therapies into clinical practice has intensified the need for robust and detailed immunophenotyping of ex vivo expanded T cells, especially considering the phenotype of CAR-T cells can predict therapeutic response and toxicity [5]. To fill this gap, we have developed and optimized an extended 36-color flow cytometry panel for deep immunophenotyping of both peripheral blood and ex vivo expanded human T cells using the Cytex Aurora full-spectrum flow cytometer (Table 1). The staining reagents required for the performance of the 36-color panel are outlined in Table 2.

We have included a PD-1 receptor occupancy (RO) component suitable for quantification of T cells bound to antibodies from immune checkpoint inhibitors (ICI) nivolumab and pembrolizumab, which are clinically approved for treatment

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TABLE 1 | Summary table.

Purpose	This panel is designed for in-depth immunophenotyping of both peripheral blood and ex vivo expanded human T cells, covering a full range of T cell fate, activation, senescence, exhaustion and functional states, with a PD-1 receptor occupancy component enabling quantification of T cells bound to antibodies from checkpoint inhibitor therapies.
Species	Human
Cell type	PBMC and ex vivo expanded T cells
Cross-references	OMIP-060, OMIP-067, OMIP-071, OMIP-075, OMIP-084, OMIP-091, OMIP-096, OMIP-099 and OMIP-114

of various cancer indications [6, 7]. The ability to monitor the occupancy of the PD-1 receptor following treatment could provide insights into therapeutic efficacy and aid in patient stratification. Moreover, ICI are often prior lines of treatment for adoptive T cell therapies. Our panel includes two clones targeting distinct PD-1 epitopes, one competitive and one non-competitive with nivolumab and pembrolizumab, which can be used to determine frequencies of bound and free receptor (Figure S17).

An example analysis using the 36-color panel is illustrated in Figure 1. Live T cells are identified using FSC/SSC, Live/Dead Blue and CD3 (Figure 1A). Gating the CD3⁺ population is performed against CD45RA using a skewed quadrant gate to exclude potential false positive events arising from spillover spreading error (Figure S10). Monitoring PD-1 receptor occupancy can be achieved by comparing staining for the competitive and noncompetitive anti-PD-1 clones with an example from a proof-of-concept study using healthy peripheral blood mononuclear cells (PBMC) shown in Figure 1B.

To define T cell subpopulations present in either peripheral blood or ex vivo expanded T cells, CD4⁺, and CD8⁺ T cells are initially selected from the CD3⁺ population, and their memory subsets identified based on the expression of CD45RA, CCR7, CD95, CD28, and CD27 [8–10] (Figure 1C). Within the CD4⁺ population, regulatory T cells (Tregs) are identified using FoxP3 and CD25 and further characterized by CD45RA to identify naïve/resting (CD45RA⁺) Tregs, and the functional marker CD39 to identify Tregs with increased immunosuppressive capacity [11, 12]. Helper T cell subsets are identified using CXCR3, CCR4, and CCR6 for Th1, Th2, and Th17 populations [8, 9] and CXCR5 for follicular helper T cells (Tfh) [13]. Expression of granzyme B within CD4⁺ cells identifies CD4 cytotoxic T lymphocytes (CTL) [14].

The PE channel is designated for T cell receptor (TCR) staining, either by pHLA-dextramer (Figure S6), or by anti-TCR antibody such as commercially available anti-V α or -V β clones. Dextramer staining allows identification of CD4⁺ and CD8⁺ T cells expressing antigen-specific TCRs, with expression in

ex vivo expanded T cells shown in Figure 1C. Both pHLA-dextramer and anti-TCR antibody detection can be used to identify transduced T cells and quantify transduction efficiency in adoptive T cell therapies.

Due to the complexity of the data produced by this 36-color panel, an analysis workflow containing dimensionality reduction and clustering algorithms is important to unlock the full potential of the data. To demonstrate the applicability of data generated by this panel in such a workflow, two datasets, one containing four healthy donor PBMC samples and the other containing four healthy donor expanded T cell samples, were analyzed separately.

Before being subject to automated analysis approaches, data were put through a clean-up process to ensure data quality. Accurate unmixing and scaling of the data were checked before the FlowAI cleaning algorithm [15] was applied to remove anomalous and artefactual events. Post-cleaning, CD3⁺ T cells were selected as the population of interest to be analyzed. An opt-SNE [16] was run using the key lineage markers for each dataset to group together events based on phenotypic similarities. The FlowSOM algorithm [17] was then run using the same markers and the resulting clusters overlaid onto the opt-SNE parameters (Figure 1D). Clusters were identified through a clustered heatmap analysis (Figure S16) and corresponded with the expected populations for both the peripheral blood and expanded T cell datasets.

Expression of individual markers associated with T cell activation (CXCR3, CD25, CD38, and HLA-DR), co-stimulation (CD226, 4-1BB, ICOS, and OX40), function (granzyme B and Ki-67), exhaustion/inhibition (CD101, CTLA-4, LAG-3, PD-1, TIGIT, and TIM-3), senescence (CD57 and KLRG1), and differentiation (CD62L and CD127) within peripheral blood and expanded T cells is illustrated in Figure 1E.

Examination of activation marker expression can provide insights into the dynamics of T cell responses in peripheral blood or expanded T cell settings, and we have therefore included several T cell activation markers spanning early to late expression kinetics. HLA-DR is a late T cell activation marker frequently used in combination with CD38 to identify activated T cell subsets [9]. A recent study found HLA-DR⁺ CD38^{hi} CD8⁺ T cells associated with systemic inflammation, tissue injury and immune disorders of severe COVID-19 patients [18]. TNFR family members OX40 and 4-1BB are transiently expressed activation markers absent in peripheral blood T cells but expressed among expanded T cells [19]. The co-stimulatory receptor ICOS regulates T cell activation and expression can be detected in peripheral blood T cells, with enhanced expression among expanded T cells [20]. CD226 is a co-activating receptor critical for CD8 T cell effector function and its axis includes competition with inhibitory receptors such as TIGIT [21]. Reduced expression has been linked to curbed effector CD8 T cell responses which could limit the efficacy of ICIs [22].

Our panel enables functional T cell characterization via intracellular staining. Granzyme B is a marker of cytotoxic function expressed highly in CD8 T cell subsets and CD4 CTLs [14]. Expression can be detected in subsets of resting T cells and

TABLE 2 | List of staining reagents comprising the 36-color panel.

Fluorochrome	Specificity	Clone	RRID	Purpose
Spark UV 387	CD3	OKT3	RRID:AB_3068126	Lineage
BUV395	CD45RA	5H9	RRID:AB_3685102	Differentiation
BUV496	OX40	ACT35	RRID:AB_2874774	Co-stimulation; activation
BUV563	Ki-67	SolA15	RRID:AB_2925398	Function
BUV615	CCR4	1G1	RRID:AB_2870269	Differentiation
BUV661	LAG-3	3DS223H	RRID:AB_2925460	Exhaustion/inhibition
BUV737	CTLA-4	14D3	RRID:AB_2895997	Exhaustion/inhibition
BUV805	ICOS	DX29	RRID:AB_2873306	Co-stimulation; activation
BV421	CD226	11A8	RRID:AB_2916829	Co-stimulation
Pacific Blue	CD27	M-T271	RRID:AB_2562505	Co-stimulation; differentiation
BV480	PD-1	EH12.1	RRID:AB_2739514	Exhaustion/inhibition; RO assay
Spark Violet 538	CD4	SK3	RRID:AB_2890774	Lineage
BV605	CD25	BC96	RRID:AB_11218989	Activation; differentiation
Super Bright 645	CCR7	3D12	RRID:AB_2744870	Differentiation
BV711	CCR6	G034E3	RRID:AB_2629608	Differentiation
BV750	CD28	CD28.2	RRID:AB_2872034	Co-stimulation; differentiation
BV785	KLRG1	2F1/KLRG1	RRID:AB_2629749	Senescence
BB515	TIM-3	7D3	RRID:AB_2744368	Exhaustion/inhibition
RB545	CD62L	DREG-56	RRID:AB_3688730	Differentiation
RB613	CD8	RPA-T8	RRID:AB_3686223	Lineage
PerCP	CD57	HI57a	RRID:AB_3718098	Senescence
RB705	CXCR5	RF8B2	RRID:AB_3689279	Differentiation
RB744	CD101	V7.1	RRID:AB_3689871	Exhaustion/inhibition
PerCP-Fire 806	4-1BB	4B4-1	RRID:AB_2941482	Co-stimulation; Activation
PE	TCR	N/A	N/A	TCR detection
Spark YG 581	CD38	S17015F	RRID:AB_2890838	Activation
PE-Dazzle 594	CXCR3	G025H7	RRID:AB_2564288	Activation; differentiation
PE-Fire 640	CD95	DX2	RRID:AB_2892365	Differentiation
PE-Fire 744	CD39	A1	RRID:AB_3106099	Exhaustion/inhibition
PE-Cy7	TIGIT	A15153G	RRID:AB_2632929	Exhaustion/inhibition
APC	PD-1	MIH4	RRID:AB_10718533	Exhaustion/inhibition; RO assay
Spark NIR 685	FoxP3	206D	RRID:AB_2890753	Differentiation
Spark Red 718	CD127	A019D5	RRID:AB_2924551	Differentiation
APC-Fire 750	Granzyme B	QA16A02	RRID:AB_2728377	Function
APC-Fire 810	HLA-DR	L243	RRID:AB_2876603	Activation
Live/Dead Blue	Viability	N/A	N/A	Viability

upregulation has been associated with fatal COVID-19 [23]. Ki-67 is another functional marker that is used to track cell proliferation kinetics [24]. Also included in the panel are several

markers of T cell exhaustion. TIM-3, LAG-3, PD-1, and TIGIT are frequently used in combinations to identify exhausted T cells, and their co-expression correlates with ICI resistance

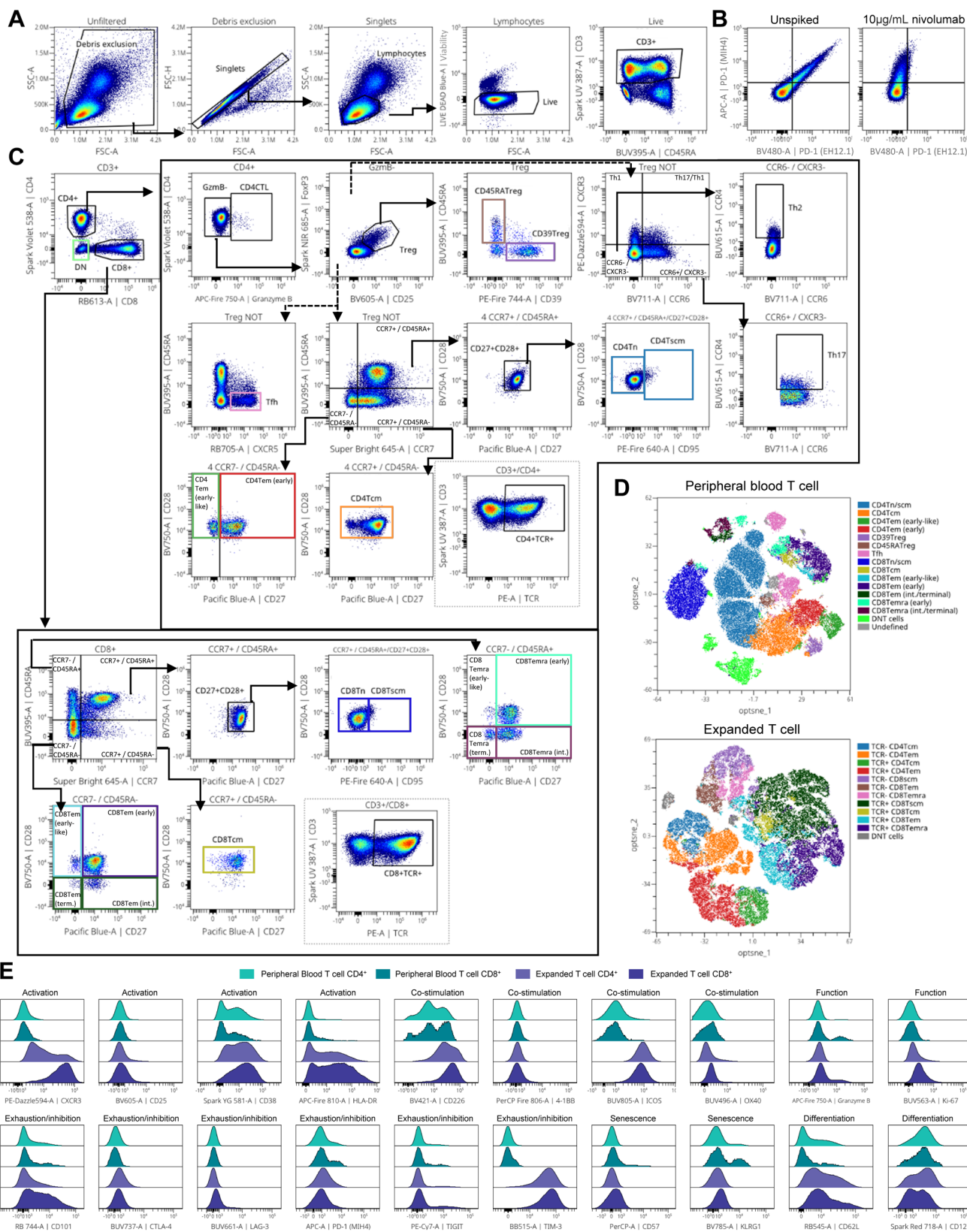


FIGURE 1 | Legend on next page.

[25]. In the field of adoptive therapies, enhanced frequencies of CAR-T cells expressing CD101 post-infusion have been shown to associate with progressive disease [5]. CTLA-4 has been shown to suppress the efficacy of CAR-T cell therapy via interaction with CD28 [26].

We have included CD57 and KLRG1 to identify terminally differentiated and senescent cell subsets. Expression of these markers positively correlates with the number of cell divisions and inversely with telomere length, and associates with reduced proliferative capacity and functionality [27, 28]. We observed a total

FIGURE 1 | Representative analysis using the 36-color panel. (A) After exclusion of debris and doublets, lymphocytes are gated based on FSC and SSC properties before selection of live and CD3⁺ T cells. Representative data from healthy PBMC. (B) Representative plots demonstrating detection of PD-1 receptor occupancy through spiking a healthy PBMC sample with nivolumab. MIH4 is noncompetitive while EH12.1 is competitive for nivolumab. Populations are pre-gated on live single lymphocytes. (C) Gating strategy for identification of T cell subsets. CD4⁺ T cells are gated and CD4⁺ CTL (GzmB⁺) and Tregs (CD25⁺ FoxP3⁺) are identified. Tregs are further characterized based on expression of CD45RA and CD39 (CD45RA⁺ CD39⁻, and CD45RA⁻ CD39⁺). Follicular helper T cells are identified from the non-Tregs (CD25⁻ FoxP3⁻) by expression of CXCR5 (CXCR5⁺ CD45RA⁻) and other helper subsets characterized from expression of CXCR3, CCR6 and CCR4: Th1 as CCR6⁻ CXCR3⁺, Th17/Th1 as CCR6⁺ CXCR3⁺, Th17 as CCR6⁺ CXCR3⁻ CCR4⁺, and Th2 as CCR6⁻ CXCR3⁻ CCR4⁺. Non-Treg cells are also further classified into memory subsets using CD45RA, CCR7, CD27, CD28, and CD95: naïve (Tn; CCR7⁺ CD45RA⁺ CD27⁺ CD28⁺ CD95⁻), stem cell memory (Tscm; CCR7⁺ CD45RA⁺ CD27⁺ CD28⁺ CD95⁺), central memory (Tcm; CCR7⁺ CD45RA⁻ CD27⁺ CD28⁺ CD95⁻), and effector memory (Tem; CCR7⁻ CD45RA⁻). CD4 Tem are further split between early (CD28⁺ CD27⁺) and early-like (CD28⁺ CD27⁻) phenotypes. CD8⁺ T cells are gated and classified into memory subsets using CD45RA, CCR7, CD27, CD28, and CD95 in concordance with the CD4⁺ non-Treg population. An additional memory subset is gated for the CD8⁺ population representing effector memory cells re-expressing CD45RA (Temra; CCR7⁻ CD45RA⁺). Both CD8⁺ Tem and Temra are subsetted into early, early-like, intermediate, and terminal states according to CD28 and CD27 expression. Double negative (DN) T cells are gated separately as CD4⁻ CD8⁻. Representative data from healthy PBMC. Gray boxes show detection of antigen-specific TCR in CD4⁺ and CD8⁺ populations for a representative expanded T cell sample. (D) FlowSOM clusters manually annotated to relevant T cell subpopulations and overlaid onto opt-SNE plots for PBMC and expanded T cell datasets. Overlays show concatenated events from four samples for each dataset. (E) Expression of individual T cell activation, co-stimulation, function, exhaustion/inhibition, senescence, and differentiation markers in peripheral blood and expanded T cell CD4 and CD8 subsets. [Color figure can be viewed at <https://onlinelibrary.wiley.com>]

loss of expression in expanded T cells, consistent with findings from anti-CD3/CD28 stimulated T cells in the literature [29]. CD62L is included in the panel to add additional power to cell fate specification and dimensionality reduction and clustering algorithms [30], while CD127 can differentiate short-lived effector cells (SLEC) from memory precursor effector cells (MPEC) in combination with KLRG1 [31].

In conclusion, we have generated a fully optimized and comprehensive T cell profiling panel designed to support advanced immunological assessments in the context of immunotherapies, including both ICI and adoptive T cell therapies.

2 | Similarity to Other OMIPs

Our panel builds upon the 31-color full-spectrum flow cytometry panel OMIP-099 with 24 shared markers, 14 of which carry the same fluorophore. Our panel also shares substantial homology (24 markers) with the 36-color full-spectrum panel OMIP-114. These panels were designed for in-depth T cell immunophenotyping within human PBMC (OMIP-099) and whole blood (OMIP-114). To date, no OMIPs have been published that are dually optimized for both peripheral blood and ex vivo expanded T cells. In addition, no published OMIPs have included an RO component such as the PD-1 RO assay included in our panel. Other human T cell focused OMIPs using full-spectrum flow cytometry with > 10 shared markers include OMIP-060, OMIP-067, OMIP-071, OMIP-075, OMIP-084, OMIP-091, OMIP-096, and OMIP-116. None of the similar OMIPs include the marker CD101, an emerging T cell exhaustion marker shown to correlate with progression following CAR-T cell therapy [5].

Author Contributions

Robert J. Harris and **Benjamin Arkle**: writing – original draft preparation, conceptualization, data curation, formal analysis, investigation, methodology, and writing – review and editing. **Elizabeth Evans**,

Rebecca Teevan and **Charles H. V. Smith**: conceptualization, data curation, formal analysis, investigation, methodology, and writing – review and editing. **Natalie Bath**: supervision, project administration, conceptualization, data curation, formal analysis, investigation, methodology, and writing – review and editing.

Acknowledgments

The authors would like to acknowledge Alex Tipping, Roger McGilvray, Christopher Dupont, Jessica Tucci-Eckstrom and the wider Translational Sciences group for their valuable comments and discussions. We would also like to thank Adam Davison from Cytex Biosciences Inc. for his helpful advice on using the SpectroFlo Autofluorescence Explorer tool for multiple AF extraction.

Funding

The authors have nothing to report.

Ethics Statement

PBMC were obtained from leukopaks purchased from HemaCare (Van Nuys, CA, USA), where all donors provided written informed consent in accordance with the Declaration of Helsinki.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.16530390>, reference number: 16530390 [32].

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** cytoa70001-sup-0001-SupinfoS1.zip. **Data S2:** cytoa70001-sup-0002-SupinfoS2.doc.